A13485H1

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B6. CONDENSED-MATTER PHYSICS

TRINITY TERM 2025

11th June, 2.30 pm – 4.30 pm

Answer **two** *questions.*

Start the answer to each question in a **fresh book.**

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

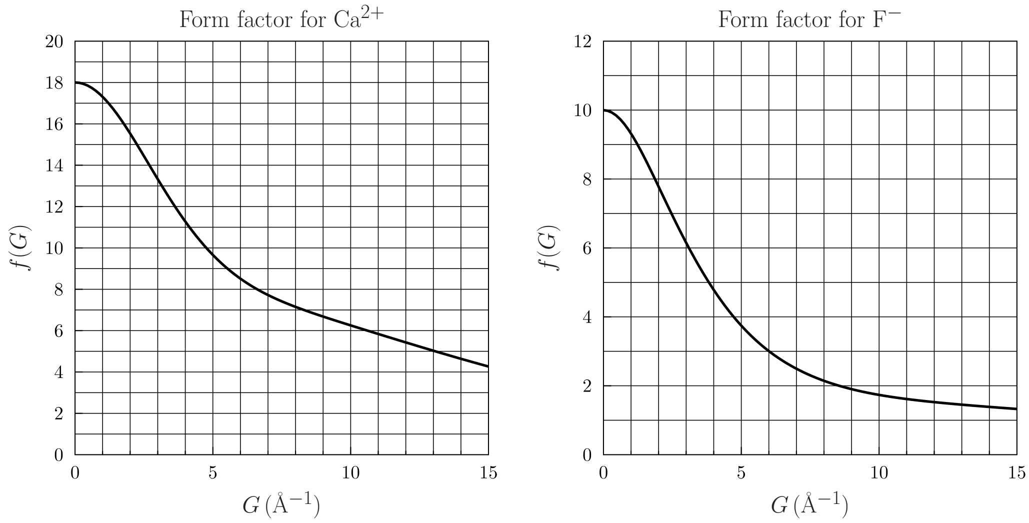
Do NOT turn over until told that you may do so.

1. (a) Explain what is meant by the structure factor $S(\mathbf{G})$ and how it is used in the determination of crystal structures using X-ray diffraction. State how the intensity of a Bragg reflection from planes with Miller indices hkl is related to the structure factor $S(\mathbf{G})$ and how the Miller indices hkl are related to the reciprocal lattice vector $\mathbf{G}$ in a cubic crystal with lattice constant a. How is $\mathbf{G}$ related to the scattering process? [6]

(b) CaF_2 crystallises with a face-centred cubic lattice with Ca ($Z = 20$) at 000 and F ($Z = 9$) at $\frac{1}{4}\frac{1}{4}\frac{1}{4}$ and $\frac{3}{4}\frac{1}{4}\frac{1}{4}$. Derive an expression for the structure factor S_{hkl} for CaF_2 in terms of the atomic form factors f_{Ca} and f_{F} for Ca and F respectively. [6]

(c) In an experiment, X-rays with wavelength 0.15406 nm are scattered from a powder sample of CaF_2. Explain why you would expect the 002 reflection to be much weaker than the 111 reflection. The first two strong reflections are found at scattering angles of 28.27° and 47.00°. Identify the Miller indices for these reflections, and deduce the lattice constant (the size of the conventional unit cell) of CaF_2. [7]

(d) The figure shows the atomic form factors of the two ions inside CaF_2. Explain the general features of these graphs. Use these data to estimate the ratio of the intensities of the first two strong reflections identified in (c). [You can ignore the Lorentz and polarisation factors.] [6]



2. (a) Consider a one-dimensional tight-binding model of electrons hopping in a crystal of identical atoms. A localised state on the nth atom is written $|n\rangle$, the lattice constant is a, the length of the crystal is $L = Na$, and we can assume that $\langle n|m\rangle = \delta_{nm}$. The only non-zero elements of the Hamiltonian $\hat{H}$ are $\langle n|\hat{H}|n\rangle = \mathcal{E}_0$ and $\langle n+1|\hat{H}|n\rangle = \langle n|\hat{H}|n+1\rangle = -t$, where t is a positive constant. You may assume periodic boundary conditions, so that e.g. $\langle 0|\hat{H}|N\rangle = -t$. Show that the energy eigenvalues for electrons in this crystal take the form

$$E(k) = A + B\cos ka,$$

and find the constants A and B. Calculate the bandwidth. [5]

Show that the density of states $g(E)$ takes the form

$$g(E) = \frac{\alpha N}{\sqrt{1-\beta(E-\mathcal{E}_0)^2}},$$

where α and β are constants which you should determine. Sketch $E(k)$ and $g(E)$. [6]

(b) A monovalent, monatomic compound can be modelled using the one-dimensional tight-binding model such that there is one electron per atom. Show that the molar heat capacity C should take the form

$$C = \eta R\left(\frac{k_{\mathrm{B}}T}{t}\right)^{\xi},$$

where η is a numerical constant that you do not need to determine, and ξ is a numerical constant that you should determine. Derive an expression for the molar magnetic susceptibility χ. [6]

How would C and χ differ if the atoms were divalent? [2]

(c) We now alter the one-dimensional tight-binding model by including *additional* non-zero elements of $\hat{H}$, which are $\langle n+2|\hat{H}|n\rangle = \langle n|\hat{H}|n+2\rangle = -t/2$. Derive an expression for $E(k)$, sketch it, and without detailed calculation sketch $g(E)$. [6]

3. (a) Explain the terms *intrinsic semiconductor* and *extrinsic semiconductor*, stating where the carriers originate from in each case. [4]

(b) Show that $n(T)$, the number of electrons per unit volume in the conduction band of a pure semiconductor at a temperature T, is given by

$$n(T) \propto T^{\alpha} \exp\left(-\frac{\eta E_{\mathrm{g}}}{k_{\mathrm{B}}T}\right),$$

where E_{g} is the energy gap and α and η are numerical constants which you should determine. What is p, the number of holes per unit volume in the valence band? [7]

(c) Define the Hall coefficient R_{H} and describe in detail how it would be measured in an experiment. Find an expression for R_{H} for a semiconductor with only one type of carrier with charge q and number density n. A sample of undoped silicon shows that R_{H} decreases by a factor of 28 when the temperature is increased from 300 to 350 K. Deduce a value of E_{g}, giving your answer in eV, and explain why it was valid to assume that predominantly electrons contribute to R_{H} and not holes. [7]

(d) A second silicon sample is lightly doped with phosphorous atoms (the atomic numbers of silicon and phosphorous are 14 and 15, respectively). Estimate the Bohr radius of an electron orbiting a phosphorous atom, and the energy needed to remove this electron from its orbit, given that silicon has relative permittivity $\epsilon_{\mathrm{r}} = 11.7$ and you may assume that electrons in silicon have an isotropic effective mass of $1.09\, m_{\mathrm{e}}$. [4]

(e) Draw a labelled sketch of $\ln R_{\mathrm{H}}$ against $1/T$ for the second silicon sample, indicating the intrinsic and extrinsic regimes. [The number density of phosphorous atoms is 6.25×10^{18} per m^3.] [3]

4. (a) What are the assumptions that go into the Einstein model and the Debye model of the heat capacity of a solid? Identify the assumptions that are common to both models and the ones that are different. [6]

(b) Using the Debye model, derive an expression for the internal energy $U(T)$ and molar low-temperature heat capacity $C_{\text{Debye}}(T)$ of a crystal containing p atoms per formula unit, showing that $C_{\text{Debye}}(T)$ takes the form

$$C_{\text{Debye}}(T) = xpR\left(\frac{T}{T_{\text{D}}}\right)^{y},$$

where R is the gas constant, T_{D} is a characteristic temperature which you should determine, and x and y are numerical constants which you should also determine. [7]

$$\left[\text{You may use: } \int_0^\infty \mathrm{d}x\,\frac{x^3}{\mathrm{e}^x - 1} = \frac{\pi^4}{15}\right]$$

(c) Show that in a crystal of a paramagnetic compound, containing N non-interacting $S = \frac{1}{2}$ moments in a volume V, and placed in a magnetic field B and held at temperature T, the partition function is $Z = [2\cosh(z)]^N$, where

$$z = \frac{\mu_{\text{B}} B}{k_{\text{B}} T}.$$

Hence derive the magnetic susceptibility $\chi(T)$ and show that it is proportional to $N/(VT)$. Find $C_{\text{mag}}(T, B)$, the magnetic contribution to the heat capacity per mole of magnetic moments, and show that it takes the form $C_{\text{mag}}(T, B) = R[f(z)]^2$, where $f(z)$ is a function you should determine. [6]

(d) A crystal ($T_{\text{D}} = 200\,\text{K}$) contains $p = 10$ atoms per formula unit, one atom of which is magnetic with $S = \frac{1}{2}$. The magnetic atoms do not interact with each other. Calculate

$$\frac{C_{\text{mag}}(T, B)}{C_{\text{Debye}}(T)},$$

the ratio of the magnetic and non-magnetic contributions to the heat capacity, at 10 K in a magnetic field of (a) 0.1 T and (b) 10 T. [3]

If the $S = \frac{1}{2}$ moments were strongly interacting, how would you expect this to change the magnetic contribution to $C_{\text{mag}}(T, 0)$? [3]